

Research Protocol v1.3

Research Title

Digital Transformation and Its Relationship with the Quality and Efficiency of Higher Education: An Explanatory and Comparative Case Study of the Universidad Nacional Mayor de San Marcos

1. Project Title

Project Description

This research investigates the relationship between digital transformation maturity, educational quality, and institutional efficiency at the Universidad Nacional Mayor de San Marcos (UNMSM). Digital transformation is conceptualized as a strategic organizational capability that integrates digital governance, technological infrastructure, leadership, organizational culture, innovation in teaching and learning, and evidence-based decision-making. Rather than focusing exclusively on technology adoption, the project examines how these dimensions are associated with differences in institutional performance across organizational units.

The project is designed as a pragmatic, explanatory and comparative mixed-methods case study capable of producing empirical evidence on associations and differences across organizational units, informing institutional modernization while contributing to international research on digital transformation in public higher education.

2. Problem Statement

Universities worldwide are undergoing rapid digital transformation in response to technological advances, changing student expectations, increasing competition, and new quality assurance requirements. Digital technologies are transforming teaching, research, administration, and institutional governance. Nevertheless, technological investment alone does not guarantee improvements in educational quality or organizational efficiency.

In many public universities, digital transformation initiatives remain fragmented and are often evaluated through indicators of technology adoption rather than organizational performance. Consequently, university leaders frequently lack empirical evidence explaining whether digital transformation genuinely improves educational outcomes and institutional effectiveness.

Within the Peruvian higher education system, public universities are implementing digital modernization strategies aligned with national policies on digital government and educational quality. However, systematic empirical studies examining the relationship between digital transformation, educational quality, and institutional efficiency remain limited. Existing research tends to emphasize descriptive analyses rather than explanatory models capable of identifying the organizational mechanisms through which digital transformation creates value.

The Universidad Nacional Mayor de San Marcos represents a highly relevant institutional context because of its historical leadership, organizational complexity, and strategic role in Peruvian higher education. Investigating digital transformation within this institution offers an opportunity to generate evidence capable of informing institutional policy while contributing to broader discussions regarding digital transformation in emerging economies.

3. Research Questions

Main Research Question

How is digital transformation maturity associated with the quality and efficiency of higher education across organizational units at the Universidad Nacional Mayor de San Marcos, and what organizational and technological factors help explain these relationships?

Supporting Research Questions

1. Which dimensions of digital transformation maturity are most strongly associated with educational quality across organizational units?
2. How is digital transformation maturity associated with institutional efficiency across academic and administrative processes and organizational units?
3. Which organizational, technological, and human factors facilitate or hinder digital transformation across organizational units?
4. How can empirical evidence generated by this research support strategic decision-making for sustainable digital transformation in public universities?

4. Research Objectives

General Objective

To examine the associations between digital transformation maturity, educational quality, and institutional efficiency across organizational units at the Universidad Nacional Mayor de San Marcos.

Specific Objectives

- Measure the maturity of digital transformation within the university.
- Determine the relationship between digital transformation and educational quality.
- Assess the relationship between digital transformation maturity and institutional efficiency across organizational units.
- Identify organizational capabilities that facilitate digital transformation.
- Develop an evidence-based framework for strengthening digital transformation in public higher education.

5. Literature Review Summary

Recent literature conceptualizes digital transformation as an organizational process rather than a purely technological initiative. Contemporary theoretical perspectives emphasize that successful digital transformation depends on leadership, governance, digital competencies, organizational culture, technological infrastructure, and institutional adaptability.

Research consistently reports positive associations between digital transformation and improvements in educational quality, operational efficiency, student engagement, organizational resilience, and institutional competitiveness. Nevertheless, significant gaps remain regarding the empirical evaluation of these relationships in public universities located in emerging economies.

Current studies also reveal limited integration between educational quality and institutional efficiency within comprehensive analytical frameworks. Most investigations examine these constructs independently, creating an opportunity for this research to contribute an integrated framework for examining the relationships between digital transformation maturity, educational quality, and institutional efficiency.

6. Explanatory and Comparative Mixed-Methods Case Study Design

The study employs an Explanatory and Comparative Mixed-Methods Case Study design within the institutional context of UNMSM.

Academic and administrative units exhibiting different naturally occurring levels of digital transformation maturity will be compared. The study does not assume experimental treatment assignment or a strong causal counterfactual. Instead, it uses existing organizational differences to assess associations between digital transformation maturity, educational quality, and institutional

efficiency while recognizing potential selection and confounding. The endogeneity, reverse-causality, selection, and confounding risks associated with these naturally occurring differences are addressed explicitly in Section 8.

Quantitative evidence will include institutional performance indicators and standardized survey instruments. Qualitative evidence will be obtained through semi-structured interviews, documentary analysis, and institutional observations. Methodological triangulation will integrate both forms of evidence to examine observed associations and differences and to explore the organizational and technological factors that may help explain them. Findings will therefore be interpreted primarily as associative and explanatory-comparative rather than as definitive causal effects. The quantitative measures will follow the construct-to-variable mapping specified in Section 7 so that survey constructs and pipeline features remain analytically consistent. The distinction between individual units of observation and organizational units of analysis, together with the planned handling of the nested data structure, is specified in Section 9.

7. Operationalization of the Main Constructs

To ensure that the research questions, instruments, institutional indicators, and reproducible analytical pipeline represent the same conceptual framework, the study will explicitly operationalize digital transformation maturity, educational quality, and institutional efficiency. Each construct will be mapped to dimensions, measurable indicators, survey or institutional measures, and corresponding dataset features.

Construct	Dimensions	Indicators	Measurement / Instrument	Data Source	Pipeline Features
Digital Transformation Maturity	Digital infrastructure; governance; process automation; organizational digital capability.	Infrastructure readiness; governance practices; automation; digital capability indicators	Validated/selected DT maturity instrument and survey items; finalized after expert validation	Faculty/staff surveys; institutional records/documents	digital_infrastructure_score ; governance_score; automation_score; related DT variables
Educational Quality	Teaching and learning quality; academic services; stakeholder experience	Teaching/learning support; academic service quality; satisfaction/experience indicators	Validated/selected educational-quality measures and institutional indicators; exact items specified in final instrument	Student/faculty surveys; institutional quality indicators	Quality-related outcome variables/features documented in data dictionary
Institutional Efficiency	Academic and administrative process efficiency; service/resource performance	Processing time; service performance; resource utilization; efficiency perceptions	Administrative performance indicators and validated/selected survey measures; finalized during instrument validation	Administrative records; staff/user surveys; institutional indicators	Efficiency-related outcome variables/features documented in data dictionary

Measurement traceability: Construct -> Dimension -> Indicator -> Survey/Institutional Measure -> Dataset Feature -> Analysis.

The exact validated instrument or maturity model, scoring procedure, item wording, and variable coding will be documented in the finalized protocol and data dictionary after instrument selection, expert validation, and pilot testing. This avoids treating provisional pipeline variables as if they were already validated empirical measures.

8. Endogeneity and Threats to Internal Validity

A key methodological concern is potential endogeneity between digital transformation maturity and institutional performance. The relationship may be reciprocal: organizational units with greater digital maturity may exhibit higher educational quality and institutional efficiency, while better-performing units may also possess stronger leadership, resources, governance capacity, and organizational readiness that enable greater adoption of digital technologies. Therefore, observed associations will not be interpreted automatically as causal effects.

Potential threats include reverse causality, selection, and confounding. Selection may arise because organizational units differ systematically in resources, leadership, staff capabilities, size, disciplinary context, and prior performance. Confounding may occur when these factors influence both digital transformation maturity and quality or efficiency outcomes. The quantitative strand will therefore document and, where data permit, adjust for relevant covariates and conduct sensitivity or robustness analyses. These procedures are intended to strengthen interpretation of associations rather than establish a definitive causal counterfactual.

The qualitative strand will complement the quantitative analysis through interviews, documentary analysis, and institutional observations. It will be used to probe plausible organizational and technological mechanisms, identify contextual explanations for observed differences, and examine alternative explanations. Integration of both strands will support a more credible explanatory and comparative interpretation while explicitly acknowledging the limits of causal inference in a single-institution case study.

9. Unit of Analysis and Multilevel Structure

The study distinguishes explicitly between the unit of observation and the unit of analysis. Students, faculty members, administrative staff, and university managers constitute the units of observation because they provide individual-level survey and

qualitative information. The principal comparative unit of analysis is the organizational unit within UNMSM, including academic and administrative units, because the research compares units characterized by different naturally occurring levels of digital transformation maturity.

The resulting data structure is therefore multilevel or nested. Individual respondents constitute Level 1 and are nested within organizational units at Level 2, all within the institutional context of UNMSM. This distinction is important because individual responses from the same organizational unit may share contextual characteristics and cannot automatically be treated as statistically independent when organizational-level comparisons are conducted.

The quantitative analysis will examine both within-unit and between-unit variation. Aggregation of individual responses to the organizational-unit level will be undertaken only when theoretically justified and supported by appropriate reliability/agreement diagnostics. If the number of organizational units and observations is sufficient, multilevel or hierarchical models will be considered to account for clustering. If those requirements are not met, clustered or unit-aware analytical procedures and sensitivity analyses will be used, with the limitation stated explicitly.

The qualitative strand will preserve respondent roles while situating interview, documentary, and observational evidence within the corresponding organizational context. Integration of the quantitative and qualitative strands will therefore distinguish individual perceptions from organizational-level patterns, preventing conclusions about organizational units from being inferred solely from unaggregated individual responses.

10. Data Management Plan Outline

The research adopts a comprehensive Data Management Plan consistent with FAIR principles.

Primary data will be collected through questionnaires, interviews, and focus groups involving academic staff, administrative personnel, university managers, and students. Secondary data will include strategic plans, institutional reports, accreditation documents, quality indicators, and digital transformation policies.

Data quality will be ensured through validation procedures, standardized coding protocols, secure digital storage, encrypted backups, and metadata documentation. Identifiable information will be accessible only to the research team, while anonymized datasets will be preserved to facilitate future scientific reuse in accordance with institutional regulations.

11. Ethical Considerations

The project will comply with internationally recognized ethical principles governing human-subject research.

Participation will be voluntary and based on informed consent. Participants will receive clear information regarding research objectives, procedures, confidentiality, risks, and expected benefits.

Special attention will be given to protecting institutional information and minimizing potential organizational risks arising from performance evaluation. The study will obtain authorization from the university and approval from the corresponding Research Ethics Committee before commencing data collection.

12. Project Schedule / Timeline

The project will be completed over a twelve-month period.

- **Months 1–2:** Comprehensive literature review and protocol refinement.
- **Month 3:** Instrument development and expert validation.
- **Month 4:** Pilot testing and methodological adjustments.
- **Months 5–6:** Quantitative and qualitative data collection.
- **Months 7–8:** Statistical analysis, qualitative coding, and methodological triangulation.
- **Month 9:** Integration and interpretation of findings.
- **Months 10–11:** Preparation of thesis chapters and scientific manuscripts.
- **Month 12:** Final revision, submission, and defense preparation.

13. Expected Results and Contributions

- The project is expected to generate robust empirical evidence on the relationships between digital transformation maturity, educational quality, and institutional efficiency within a leading public university in Peru.
- From a **theoretical perspective**, the study will contribute an integrated conceptual framework linking digital transformation, educational quality, organizational capabilities, and institutional efficiency.
- From a methodological perspective, the research will apply an explanatory and comparative mixed-methods case study framework suitable for investigating complex organizational phenomena in higher education.
- From a **practical perspective**, the findings will support evidence-based decision-making by providing strategic recommendations for strengthening

digital governance, optimizing institutional processes, and improving educational outcomes.

- Finally, from a **policy perspective**, the proposed framework is expected to be transferable to other public universities in Peru and Latin America, contributing to the development of sustainable digital transformation strategies and more effective higher education systems.